Project (E) 448 Responsibilities & Student's GA Achievement Plan Ver2.2

This form must be filled out by the student, signed and handed in to the module administrator early in the semester. Both the student and the supervisor must sign the agreement on main details and mutual responsibilities. The student should discuss their GA achievement plan with the supervisor, before completing this form. Only the student signs their GA achievement plan, as it is their own responsibility. ***This plan is NOT a guaranteed recipe for passing Project (E) 448.*** *Rather, it serves as a record of the student having considered these important aspects at an appropriately early stage. GA achievement plans should be revised as needed and in consultation with the supervisor, during the course of the project.*

Main details

Student	Initials and surname	UVS Rosant	SU number	26948982
Supervisor	Initials and surname	RP, Theart		
Project title	Local-Only Reasoning for Autonomous Rovers: Plan Execution, Report Generation, and the Limits of Command Ambiguity			
Project description, including the aim, scope and envisioned approach *(max. 150 words)*	This project investigates whether a fully local, cloud-free AI pipeline can enable an indoor rover to interpret vague natural-language voice commands, plan and execute multi-step missions, and report back autonomously, without mid-mission human intervention. The rover (a TurboPi platform with Raspberry Pi 5) streams commands and video to a laptop-hosted inference backend, where speech-to-text, a vision-language model (ambiguity checking and execution monitoring) and a reasoning large language model (mission planning) run entirely on-device. The system resolves genuine ambiguity by clarifying before departure, self-corrects during the mission, and flags residual uncertainty in a structured report afterwards, rather than halting for input mid-task. The central research question is two fold: first, whether local models can sustain this full decode-execute-report loop reliably; second, how much complexity or vagueness a command can carry before the system's reasoning breaks down or hallucinates, establishing a practical limit on sequential, ambiguous instruction handling for on-device autonomous systems.			

Mutual responsibilities

1. It is the responsibility of the student to clarify aspects such as the definition and scope of the project, the place of study, research methodology, reporting opportunities and -methods (e.g. progress reports, internal presentations and conferences) with the supervisor.
2. It is the responsibility of the supervisor to give regular guidance and feedback with regard to the literature, methodology and progress.
3. The rules regarding submission and evaluation of the project is outlined in the module framework and SUNLearn page and will be strictly adhered to.
4. The supervisor conveyed the departmental view on plagiarism to the student, and the student acknowledges the seriousness of such an offence.
5. The supervisor certifies that the project as described above has sufficient scope to achieve, in principle, the required GAs.
6. It is the responsibility of the student to initiate a discussion with the supervisor on GA achievement prior to filling out and handing in this form.

Signatures for agreement on main details and mutual responsibilities

Role	Signature	Date
Student		05/08/2026
Supervisor		05/08/2026

Student's graduate attribute (GA) achievement plan

How will GA 1 **(problem solving)** be achieved? *(<=100 words)*
The core problem is enabling a robot to interpret vague natural-language instructions and act autonomously, with no human present during the mission. This is formulated and analysed by researching literature (e.g. Introspective Planning, conformal prediction) on instruction ambiguity in large models, then testing from first principles whether these reasoning patterns hold once models are quantized for on-device deployment. Sustainable development is addressed directly through this local-only architecture: running inference on-device removes the recurring energy and data-transfer cost of cloud-hosted models, reducing the system's ongoing carbon and infrastructure footprint relative to a cloud-dependent equivalent.
How will GA 2 **(application of scientific and engineering knowledge)** be achieved? *(<=100 words)*
The project applies knowledge of computing fundamentals, embedded systems and sensor data processing (camera, ultrasonic and line sensors) to develop a working rover perception and navigation solution. Locally hosted language and vision-language models are accessed via the Ollama API rather than built or trained from scratch, so engineering knowledge is applied in model selection and evaluation, and in designing the backend service and front-end dashboard that integrate these models into a full decode-execute-report pipeline.
How will GA 3 **(engineering design)** be achieved? *(<=100 words)*
A system architecture will be designed comprising an ambiguity-check at command time, so genuinely ambiguous instructions are clarified before departure, and a self-correction mechanism during the mission that compares live observations against expected outcomes. Residual uncertainty is flagged in the final report rather than halting the mission for input. The design also integrates camera, ultrasonic and line-sensor input for navigation, and splits reasoning into a fast, continuous vision-based check and a slower language-model replanning step, meeting the need for autonomous operation on a resource-constrained on-device platform. This reuses existing hardware, avoiding added cloud infrastructure and minimising whole-life resource cost.
How will GA 4 **(investigations, experiments and data analysis)** be achieved? *(<=100 words)*

A structured test sprint will investigate two questions: whether locally hosted models can sustain the full decode-execute-report pipeline, and how much complexity or vagueness a command can carry before the system's reasoning breaks down. This uses a logged, scoreable test harness comparing candidate models (e.g. Moondream, Gemma4 E2B, qwen2.5vl:3B), following a methodology informed by the literature on introspective planning and conformal prediction. Results are analysed to reach substantiated conclusions on model and hardware viability and on a practical instruction-complexity limit.

How will GA 5 **(engineering methods, skills and tools, including IT)** be achieved? *(<=100 words)*

Modern engineering and IT tools will be used throughout: Ollama and LM Studio for serving and evaluating locally hosted models, Python and OpenCV for camera-based perception, FastAPI for the backend service, and Next.js for the live mission dashboard, running on a laptop-hosted inference backend with a Raspberry Pi 5 based TurboPi rover as the physical platform. Tool limitations will be identified and worked around in practice, such as model memory footprints and inference latency relative to the Raspberry Pi's constraints.

How will GA 6 **(professional and technical communication)** be achieved? *(<=100 words)*

The project includes a written report and an oral presentation. These demonstrate competence to communicate effectively, both orally and in writing.

How will GA 8 **(individual work)** be achieved? *(<=100 words)*

The student will take primary responsibility for successful completion of all aspects of the project.

How will GA 9 **(independent learning ability)** be achieved? *(<=100 words)*

For successful completion of the project, the student is required to acquire knowledge independently (from the literature or the internet, for example) and without the context of this required knowledge being fully specified in the project definition.

Signature acknowledging own responsibility to achieve GAs

Student	Signature	Date
		05/08/2026